

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Question Paper Name :	IIT M DEGREE AN EXAM QDB2 15 Mar 2026
Subject Name :	2026 Mar15: IIT M AN EXAM QDB2
Creation Date :	2026-03-05 10:37:28
Duration :	120
Total Marks :	637
Display Marks:	Yes
Share Answer Key With Delivery Engine :	Yes
Actual Answer Key :	Yes
Calculator :	Scientific
Magnifying Glass Required? :	No
Ruler Required? :	No
Eraser Required? :	No
Scratch Pad Required? :	No
Rough Sketch/Notepad Required? :	No
Protractor Required? :	No
Show Watermark on Console? :	Yes
Highlighter :	No
Auto Save on Console?	Yes
Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Group I

Group Number :	1
Group Id :	64065332737
Group Maximum Duration :	0
Group Minimum Duration :	90

Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	637
Is this Group for Examiner? :	No
Examiner permission :	Cant View
Show Progress Bar? :	No
Revisit allowed for group Instructions? :	Yes
Maximum Instruction Time :	0
Minimum Instruction Time :	0
Group Time In :	Minutes
Revisit Section :	Yes
Action on Revisit Section :	View and Edit
Navigate To Group Summary From Last Question? :	No
Disable Submit Button During Assessment? :	No
Section Selection Time? :	0
No of Optional sections to be attempted :	0

Industry 4.0

Section Id :	640653129736
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	15
Number of Questions to be attempted :	15
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653362946
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406531961640

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536374067.  Yes

6406536374068.  No

Sub-Section Number :

2

Sub-Section Id :

640653362947

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406531961641

Correct Marks : 2

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets in the neighbouring shelves. He picked a few milk-powder packets, however these are not part of his list. He was searching for Good-day biscuits. He could identify only one packet in the shelf remaining. However, there were lot of Sun-fist biscuits lying in the next shelf. So he picked ten Sun-fist packets. Now map the scenarios with the items below:

Product	Scenario
P1)Tea: Coffee	S1)Complementary
P2) Tea: milk-powder packet	S2)Substitution
P3) Good-day biscuits: Sun-fist bicuit	S3)Inventory dependant demand
	S4)Impulse buying

Options :

6406536374069.  P1-S1, P2-S3, P3-S4

6406536374070.  P1-S2, P2-S1, P3-S3

6406536374071.  P1-S2, P2-S1, P3-S4

Sub-Section Number :

3

Sub-Section Id :

640653362948

Question Shuffling Allowed :

No

Question Id : 6406531961642 Sub Question Shuffling Allowed : No Group Comprehension

Questions : No Question Pattern Type : NonMatrix

Question Numbers : (3 to 4)

Question Label : Comprehension

A factory works for two 8-hour shifts each day and produces a total of 30,000 electric bulbs using a set of workstations. There are 8 activities required to manufacture the bulb, such that the sum of all activity times is equal to 12 seconds.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 3 Question Id : 6406531961643

Correct Marks : 1.5

Question Label : Short Answer Question

What is the cycle time (in seconds) at any workstations?
(round your answer to one decimal)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.7 to 2.0

Question Number : 4 Question Id : 6406531961644

Correct Marks : 1.5

Question Label : Short Answer Question

What is the theoretical minimum number of workstations required?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

7

Sub-Section Number :

4

Sub-Section Id :

640653362949

Question Shuffling Allowed :

Yes

Question Number : 5 Question Id : 6406531961645

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the examples for Offshoring from the list below:

Options :

6406536374074. ✓ Micron Technology, a US based semiconductor giant, setting up a Supply

Chain Analytics team, in India.

6406536374075. ✓ Micron signs an agreement with Taiwan based TSMC Co., to manufacture silicon wafer for their ICs.

6406536374076. ✗ Western Australia's electricity network agency assigns Tata Consultancy Services at Pune, to support it with an overhaul of its legacy IT systems, aimed at delivering cheaper and more reliable power.

6406536374077. ✗ Tata Motors assigns the manufacturing of its braking system to TACO group.

Question Number : 6 Question Id : 6406531961647

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

6406536374079. ✗ Optimal solution in a facility location problem using Euclidean method will result in a region

6406536374080. ✓ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity

6406536374081. ✓ Optimal solution in a facility location problem using Euclidean method will result in a point

6406536374082. ✗ Optimal solution in a facility location problem using Euclidean method matches the center of gravity in not of the cases

Question Number : 7 Question Id : 6406531961656

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is an advantage of Process Layout?

Options :

6406536374113. ✗ Low material handling

6406536374114. ✗ Total production time per unit is small

6406536374115. ✗ Low training cost for the operators

6406536374116. ✓ It can accommodate high variety of products

Sub-Section Number :

5

Sub-Section Id :

640653362950

Question Shuffling Allowed :

Yes

Question Number : 8 Question Id : 6406531961646

Correct Marks : 3

Question Label : Short Answer Question

Rana Bakery, a company is using a layout which consists of 9 workstations. 200 sandwiches are produced in a day on this layout. A typical day consists of a single shift which runs for 10 hours. The sandwich-making process consists of 9 activities (the information is provided in the table

below).

Activity ID	Activity Description	Activity Time (in seconds)	Preceding activities
A	Wheat bag emptying	20	
B	Vegetable cleaning	10	
C	Bread baking	60	A
D	Vegetable cutting	40	B
E	Bread Slicing	20	C
F	Butter applying	5	E
G	Vegetable staffing	5	E, F
H	Sandwich baking	15	G
I	Sandwich packing	5	H

What is the current efficiency of the line where each activity is performed on a separate workstation (Enter only the numerical value up to two decimal places without the “%” symbol. Example: if the answer is 99.756%” enter only “99.76”)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

11 to 12

Sub-Section Number :

6

Sub-Section Id :

640653362951

Question Shuffling Allowed :

Yes

Question Number : 9 Question Id : 6406531961648

Correct Marks : 1

Question Label : Multiple Choice Question

The Production rate of an assembly line is determined by its

Options :

6406536374083. ✓ Slowest station

6406536374084. ✗ Fastest station

6406536374085. ✗ Idle station

6406536374086. ✗ None of these

Question Number : 10 Question Id : 6406531961649

Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following is an advantage of Product Layout?

Options :

6406536374087. ✗ It can accommodate high variety of products

6406536374088. ✓ Small in-process inventory

6406536374089. ✖ High utilization of machines

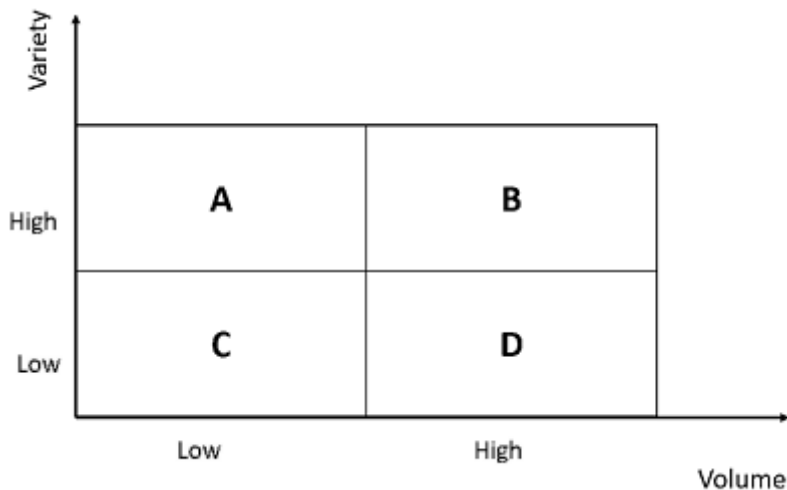
6406536374090. ✖ Low investment

Question Number : 11 Question Id : 6406531961650

Correct Marks : 1

Question Label : Multiple Choice Question

Select the appropriate layout for Quadrants A, B, D in the following figure:



Options :

6406536374091. ✖ A: Group layout, B: Product layout , D: Process layout

6406536374092. ✖ A: Product layout, B: Process layout , D: Group layout

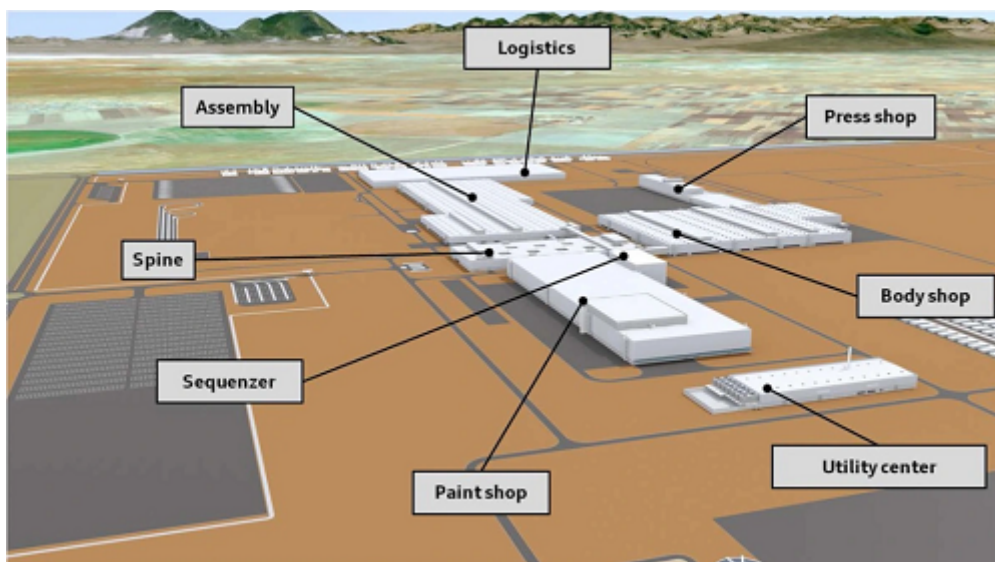
6406536374093. ✔ A: Process layout, B:Group layout , D: Product layout

Question Number : 12 Question Id : 6406531961651

Correct Marks : 1

Question Label : Multiple Choice Question

Following is the layout from an aircraft manufacturing company. Identify the layout from the figure below.



Options :

- 6406536374094. ✖ Product layout
- 6406536374095. ✖ Process Layout
- 6406536374096. ✖ Group layout
- 6406536374097. ✔ Fixed position layout.

Question Number : 13 Question Id : 6406531961652

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

- 6406536374098. ✖ IC engine
- 6406536374099. ✖ Electric car
- 6406536374100. ✔ Hyperloop

Question Number : 14 Question Id : 6406531961653

Correct Marks : 1

Question Label : Multiple Choice Question

Select the layout which do not require general supervision:

Options :

- 6406536374101. ✖ Product layout
- 6406536374102. ✔ Process layout
- 6406536374103. ✖ Group layout
- 6406536374104. ✖ Fixed position layout

Question Number : 15 Question Id : 6406531961654

Correct Marks : 1

Question Label : Multiple Choice Question

Match Col A with Col B for landmark outcome of industrial era:

Col A (Industrial Era)	Col B (Major Outcome)
1.0	Digitization
2.0	CNC
3.0	Steam Engine
4.0	Mass production

Options :

- 6406536374105. ✖ 1.0:CNC, 2.0:Mass production, 3.0: Digitization, 4.0:Steam Engine
- 6406536374106. ✖ 1.0:Steam Engine, 2.0:mass production, 3.0: digitization, 4.0:CNC
- 6406536374107. ✔ 1.0:Steam Engine, 2.0:mass production, 3.0: CNC, 4.0: Digitization
- 6406536374108. ✖ 1.0:Mass production, 2.0:Steam Engine, 3.0: digitization, 4.0:CNC

Sub-Section Number :

7

Sub-Section Id :

640653362952

Question Shuffling Allowed :

Yes

Question Number : 16 Question Id : 6406531961655

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

When will of the following scenarios, a facility layout problem is relevant?

Options :

6406536374109. ✓ A new facility is added

6406536374110. ✓ A new product is introduced

6406536374111. ✓ A new service is introduced

6406536374112. ✓ Shift in technology